a
$Li_{24}P_4S_{20}Cl_4$ (Li_6PS_5Cl), a = 10.28 Å
Li (24)
P (4)
S (20)
Cl (4)
P-S < 2.6 Å
Ordered S/Cl: Cl on 4a, free S on 4c,
the arrangement of the AIMD reference (Deng et al. 2017)
b
Structure
Li_6PS_5Cl
52 atoms (a)
416 atoms (2x2x2)
MD, universal potentials
MACE-MP-0 small (foundation)
MatterSim
52 atoms: 50, 100, 150 ps
416 atoms, 200 ps
600, 800, 1000 K
AIMD of the same ordered cell
Deng et al. 2017
52 atoms
a = 10.10 Å
600-1200 K
100 ps per T
Ext. Data Fig. 8
MACE-MP-0, 52 atoms, 50 ps
27 configurations
frames 100-900
9 per temperature
DFT labels
QE 7.1, PBE
Gamma point
2x2x2 k-points
Fine-tunes of MACE-MP-0
Gamma labels
2x2x2 labels
Cl excess
$Li_{6-x}PS_{5-x}Cl_{1+x}$
x = 0-0.75
MACE-MP-0
392-416 atoms
150 ps
Ext. Data Fig. 9
Fig. 2 Against DFT labels
forces and energies
MACE-MP-0 and both fine-tunes
held-out configurations
Fine-tuned MD
416 atoms, 200 ps
600, 800, 1000 K
Gamma: stage 2
2x2x2: stage 1
Fig. 3 Against AIMD
Li D at 600-1000 K
all four models
no extrapolation
Fig. 4 Fine-tuned against foundation
fine-tuned MD against MACE-MP-0 MD
pre-registered reading
same MD protocol
416 atoms, 200 ps, 600-1000 K
Every model is compared with DFT: forces with the DFT labels, Li diffusion with AIMD of the same ordered cell.
c
Potential or composition
600 K
800 K
1000 K
Atoms
Production (ps)
Universal potentials
MACE-MP-0 log log log 52 50
MatterSim log log log 52 50
MACE-MP-0 52 100
MatterSim log log log 52 100
MACE-MP-0 log log log 52 150
MatterSim 52 150
MACE-MP-0 log 416 200
MatterSim log 416 200
Fine-tuned
Fine-tuned, Gamma labels 416 200
Fine-tuned, 2x2x2 labels log 416 200
Cl excess
Li_6PS_5Cl (x = 0) log 416 150
$Li_{5.75}PS_{4.75}Cl_{1.25}$ (x = 0.25) log 408 150
$Li_{5.5}PS_{4.5}Cl_{1.5}$ (x = 0.50) log 400 150
$Li_{5.25}PS_{4.25}Cl_{1.75}$ (x = 0.75) log 392 150
Published in July 2026 and reproduced here (33 runs): pymatgen class R1, kinisi D and its uncertainty class R2 (Ext. Data Fig. 2)
Analysed here for the first time (9 runs)
log
July run log survives (19 runs): thermostat statistics recomputed from the trajectory are checked against it; the other 23 runs have statistics from the trajectory only
MACE-MP-0
MatterSim
Fine-tuned, Gamma labels
Fine-tuned, 2x2x2 labels
416 atoms, production
52 atoms
Cl-excess series